ICT CORNER

Ray optics

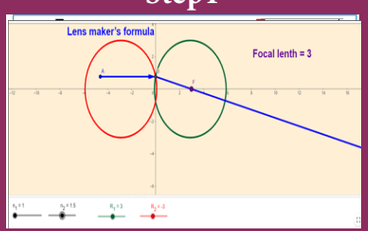
In this activity you will be able to find out the radius of curvature needed to make a lens of desired focal length

Topic: Lens maker's formula.

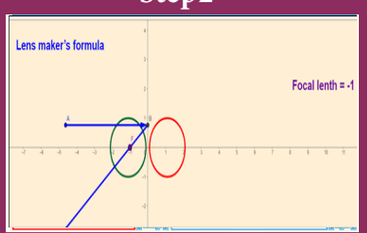
STEPS:

- In the browser go to the page "https://www.geogebra.org/m/wv7eccvc".
- Here n_1 = refractive index of air ($n_1 = 1$). n_2 is the refractive index of the lens. R_1 and R_2 are the radius of curvature of the lens.
- Take $n_2 = 1.5$, $R_1 = 3$ cm and $R_2 = -3$ cm. Calculate the focal length of the lens using the formula $\frac{1}{f} = (n_2 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$. Verify your answer with the value obtained on the screen by adjusting n_2, R_1 and R_2 .
- Find out the curvatures R_1 and R_2 to make a lens of desired focal length 7 cm.
- When $R_1 = -1$cm and $R_2 = 1$cm, you will get a concave lens of focal length −1 cm. Try to construct concave and convex lenses by adjusting R_1 and R_2.
- Adjust the refractive index n_2, and analyse how focal length changes with respect to refractive index.
- Adjust the refractive index n_1, and analyse how focal length changes with respect to refractive index of the medium surrounded by the lens

Step1



Step2



Note:

Use flash enabled browser or install flash player in your system.

URL:

https://www.geogebra.org/m/wv7ecvc

* Pictures are indicative only.

* If browser requires, allow **Flash Player** or **Java Script** to load the page.

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